



Anil Seth

The Linux Desktop Through the Decades

The author reminisces about how things were, and how much better they are now... while offering some ideas about what is yet to come.

As another year ends and a new one begins, it does not seem so long ago that we had set up a demo of networked workstations for a financial company. The consultant shot us down in one minute. He said that if the analysts felt that their spreadsheets could be accessed by their colleagues, they would not create them!

Some years later, it took a year for me to get the first copy of Linux CDs ordered from the USA. A part of the delay was because it involved such a low cost that the purchase department forgot about it! Finally, we got Slackware with Linux kernel 1.2x or 1.3x. Subsequently, it became easier to get various distributions when GT started distributing them from Bangalore.

The next major change was when magazines started offering distros and patches on CDs. Our first effort on receiving the magazines was to make sure that the CDs were readable, and apply the patches.

However, the computers were slow and had a limited amount of memory. To squeeze the most out of our desktops, we would customise the kernel and remove as much as we could. Sometimes, we needed to add some drivers by using the source code. Now, I can't remember how many years it has been since I compiled a kernel.

Network speeds improved, but magazines started distributing DVDs. We were updating desktops and servers using the Internet, but to upgrade the distribution, we still relied on the magazines, or services that would send us the DVDs.

Tools like *yum* and *apt* were not that common. We relied on search engines like *rpm.phone.net* to find pre-packaged applications for our preferred distribution. Installing them was also not always easy, as handling dependencies had to be done manually. Some enthusiastic individuals would make life easier for us by packaging and making available their preferred group of applications on their sites. If our preferred source was FreshRPMs, but we needed one package from Livna, it wasn't always a sure shot. If a dependency needed by both was packaged by both, it could result in one or more applications not working correctly. It seemed simple enough and very beneficial to us, but in retrospect, it is easy to understand why convincing friends and colleagues to switch to Linux was not very easy. Now, with the widespread use of *yum* and *apt-get*, and the consolidation of various repositories, version or packaging conflicts are very rare.

As the networking speeds improved, we could upgrade our home computers within days of the official releases. We could explore beta releases as well. On the down side, the range of software in various repositories has increased, so we install more packages—and the updates to various packages appear to have

become more frequent. Given that our broadband wasn't 'broad' enough, the introduction of delta packages by OpenSUSE and Fedora was a great boon. In fact, I regularly started using ArchLinux as well, when a user set up a delta repository for it. However, he did not need it after a while, and nothing replaced it. Now, I rarely use ArchLinux. I may look at it again as now there appears to be a delta repository for both i386 and x86_64—delta.archlinux.fr.

I had explored OpenSUSE Tumbleweed, but had given up on it as it did not have delta package support, besides a few other issues.

At college, with a much higher bandwidth, the time required to download the full package is somewhat less than the time needed to re-construct the package using delta-rpm! It is no surprise that the need for delta repositories is not that great in network-rich countries. In a couple of years, I expect that the same will be true for our home PCs as well.

What may happen next?

If an application is needed but is not installed, Ubuntu often recommends a package that may be installed. This approach may become more widespread, and maybe even transparent.

There is also the possibility that apps, as on mobile phones, may become the default way we use desktops as well. In this case, there is the possibility of one or more apps for every common website we may be accessing. The number of apps that will need to be managed may soon become ridiculously large, especially as some will be used rarely or just once.

Browser applications may give us an indication of where we are headed. Any time we use a browser, it tells us if an upgrade is available. This may be a useful step, in general. Why update packages that are not being used? Package managers can become smart enough to install or update packages and their dependencies, just in time. Or, quite possibly, we may all be using something like a ChromeBook or Firefox OS, and the current systems may slowly vanish or become irrelevant.

Whatever may happen, the next few years are bound to be exciting and full of surprises. 

By: Anil Seth

The author is currently a visiting faculty member at IIT-Ropar, prior to which he was a professor at Padre Conceicao College of Engineering (PCEC) in Goa. He has managed IT and imaging solutions for Phil Corporation (Goa), and worked for Tata Burroughs/TIL. You can find him online at <http://sethani.com/> and reach him via email at anil@sethani.com.